

Alice Ku

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EDUCATION

University of Cambridge

MPhil, Medical Science (Research), Computational Imaging

Cambridge, UK

Expected August 2026

Stanford University

M.S., Electrical Engineering

Stanford, CA

September 2021-June 2025

Concentration: Signal Processing, Control, Optimization

B.S., Engineering Physics

September 2021-June 2025

Minor: Music

Relevant coursework: Medical Imaging and Systems, Signal Processing, Linear Dynamical Systems, Computer Systems

MEDICAL IMAGING RESEARCH

Vendor-neutral MRI sequence development and reconstruction

October 2025-present

PI: Professor Martin Graves (University of Cambridge)

- Implemented non-Cartesian signal acquisition and reconstruction pipelines with Pulseseq and GE environments
- Developed an integrated MATLAB framework for vendor-neutral MRI sequence development, combining pulse sequence generation, scanner control, image reconstruction, visualization, and validation
- Integrated Pulseseq, TOPPE, GE EPIC, Orchestra SDK, and custom MATLAB software into an automated workflow to support rapid sequence prototyping across multiple software environments
- Investigated and resolved hardware timing, gradient, sampling, and reconstruction issues through quantitative analysis of k-space trajectories, system constraints, and image quality
- Performed MRI scans to debug the acquisition workflow and improve image quality

Multi-channel diffusion-model denoiser for diffusion MRI

September 2024-June 2025

PI: Professor Kawin Setsompop (Stanford University)

- Built a Python pipeline to automate multi-database image preprocessing, validation, and statistical analysis
- Developed, trained, and evaluated single-channel and multi-channel diffusion denoising models

Hallucination analysis of diffusion models for CT denoising

July 2024-September 2024

PI: Professor Dufan Wu (Harvard Medical School)

- Implemented a Python pipeline for diffusion-model-based CT denoising
- Designed quantitative evaluation methods to characterize hallucination with spatial frequency decomposition
- Published a first-author IEEE ISBI conference proceeding

Denoising X-ray images with deep learning: Impact of spatially correlated noise

June 2023-February 2024

PI: Professor Adam Wang (Stanford University)

- Developed physics-based simulation to model spatially correlated noise in X-ray imaging systems
- Trained and evaluated the denoising RED-CNN algorithm with Noise-to-Noise/Noise-to-Clean learning strategies
- Designed controlled experiments to quantify the impact of detector physics on denoising performance
- Published a first-author SPIE Medical Imaging conference proceeding and selected for oral presentation

OpenDosimeter: Low-cost, open-source X-ray dosimeter

April 2023-June 2024

PI: Professor Adam Wang (Stanford University)

- Contributed to the design of an open-source personal X-ray dosimeter based on a SiPM and Raspberry Pi Pico
- CAD-ed and 3D-printed iterations of enclosures to improve durability and usability

PUBLICATIONS

- **Ku, A.**, Tivnan, M., & Wu, D. (2025, April). Hallucination Analysis of Score-Based Diffusion Models for CT Denoising Through Spatial Frequency Decomposition. *IEEE 22nd International Symposium on Biomedical Imaging* (pp. 1-5). IEEE.

- **Ku, A.**, Wang, S., & Wang, A. (2024, April). Denoising x-ray images with deep learning: impact of spatially correlated noise. In *Medical Imaging 2024: Physics of Medical Imaging* (Vol. 12925, pp. 160-163). SPIE.
- Ger, N., **Ku, A.**, Lopez, J., Bennett, N.R., Wang, J., Ateka, G., Anyenda, E., Rosezky, M., Wang, A.S., & Shaker, K. (2025, October). OpenDosimeter: Open hardware personal X-ray dosimeter. Accepted by Communications Engineering.

WORK EXPERIENCE

Taiwan Semiconductor Manufacturing Company (TSMC)

July 2025-August 2025

Optical Proximity Correction (OPC) R&D Intern

- Developed an automated Source-Mask Optimization (SMO) framework in Python by integrating machine learning and source fusion techniques to improve hotspot imaging
- Applied K-means clustering to large-scale diffraction pattern data to automate source optimization and identify optical source topologies for critical layout features
- Analyzed the optical effects of illumination source shapes using Zernike polynomial decomposition and pixelated point-source analysis, establishing physics-based relationships between diffraction behavior and pattern printability
- Results have been adopted by the engineering team and deployed in production VB layer optimization, effectively boosting optical quality (ILS by 2% and DOF by 5%)

SKILLS

- *Programming:* Python • MATLAB • C++ • C
- *Algorithms & Engineering:* signal processing • image reconstruction • data analysis • simulation • experimental design
- *Software:* Linux • Git/GitHub • MATLAB app designer • object-oriented APIs • GE EPIC

LEADERSHIP & VOLUNTEERING

- Chair of Mentorship, Cap and Gown Women's Leadership Organization
- Tutor, Stanford PHYSICS 41 Course: Mechanics
- Tutor, Stanford Program for Inspiring the Next Generation Women in Physics